

Management of IgG4-related disease

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IgG4-related disease was unrecognised as a unified disease entity until this century, yet in a short period of time the disease has been appreciated to have a worldwide distribution, and its clinical, pathological, and radiological features have been described in considerable detail. The disease has strong organ predilections, and many of the clinical presentations of disease are increasingly familiar to both generalists and specialists. Early recognition of IgG4-related disease is crucial because although the disease is highly treatable, it can lead to serious organ damage and even death if undiagnosed until advanced stages. Its indolent nature often makes diagnosis challenging, and IgG4-related disease is one of the great mimickers of other diseases in the current era. Glucocorticoids are an effective treatment for IgG4-related disease, but their long-term use is problematic in a disease that frequently affects middle-aged to elderly individuals and often leads to pancreatic dysfunction. Our understanding of the pathophysiology of the disease is surprisingly advanced given the relatively recent recognition of this condition. Insights into disease pathophysiology offer the possibility of a variety of targeted treatment approaches. Looking ahead, biological therapies could profoundly alter the way in which IgG4-related disease is managed, permitting the use of specific therapies that are tailored to patients' clinical phenotypes.

Introduction

IgG4-related disease is a multi-organ, immune-mediated condition that was recognised as a unified disease only in 2003.^{1,2} However, clinical presentations now known to be classic features of IgG4-related disease appeared in the medical literature much earlier, with descriptions in the late 1800s of Mikulicz disease (simultaneous enlargement of the lacrimal and major salivary glands);³ Küttner's tumour (isolated submandibular gland enlargement);⁴ and Riedel's thyroiditis.⁵ Many of the other organ system manifestations now linked to IgG4-related disease were identified during the 1900s, but these were thought to be manifestations of single-organ diseases⁶ or processes restricted to single-body regions,⁷ rather than parts of a systemic inflammatory disease. The connection between clinical features of disease and elevated serum concentrations of IgG4 was established in 2000, when Hamano and colleagues⁸ reported the utility of elevated serum IgG4 concentrations in distinguishing between sclerosing pancreatitis—now known as type 1 IgG4-related autoimmune pancreatitis—and other hepatobiliary conditions.

In the past decade, there has been an extraordinary expansion in knowledge about IgG4-related disease. Furthermore, the indolent nature of the disease and the use of targeted treatment approaches have provided major insights into its pathophysiology. These insights, in turn, suggest the possibility of several mechanism-based approaches to therapy. In this review, we discuss the pathology and pathophysiology of IgG4-related disease and provide an overview of the clinical management of patients with the disease.

Pathology and pathophysiology

The organs affected by IgG4-related disease all share similar histopathological and immunostaining features.⁹ The hallmark findings include a dense lymphoplasmacytic infiltrate rich in IgG4-positive plasma cells (figure 1A). This cellular infiltrate is enmeshed in fibrosis with an

irregular, whorled appearance termed storiform fibrosis (figure 1B).¹⁰ Obliterative phlebitis and eosinophilic infiltration are observed in approximately half of patients with IgG4-related disease. By contrast, necrotising features, granulomatous inflammation, and multinucleated giant

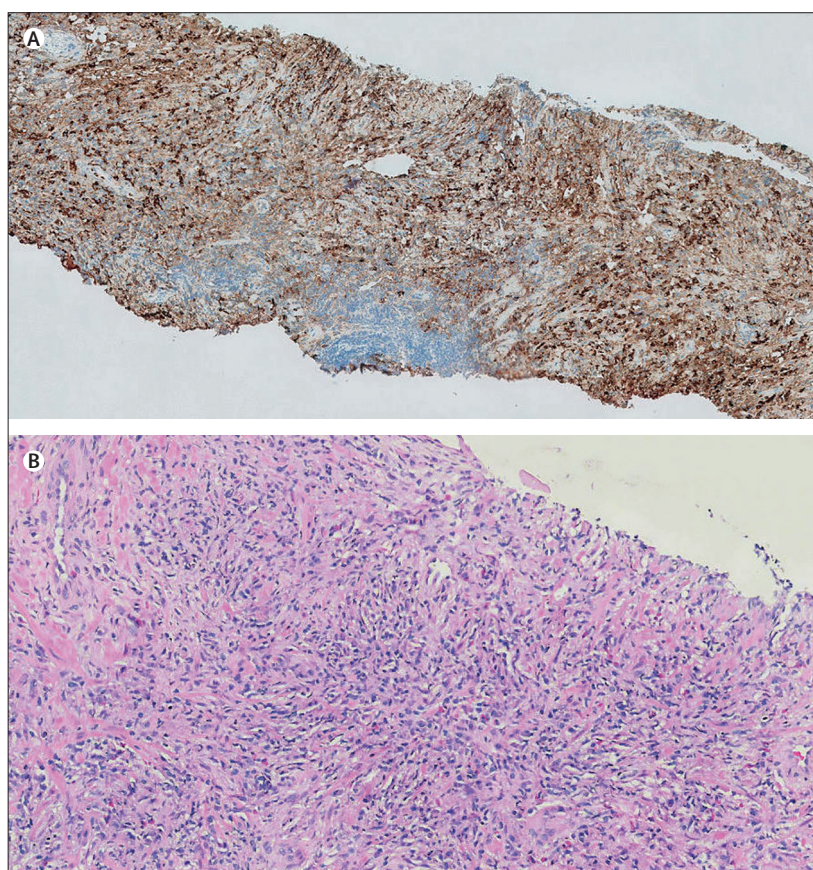


Figure 1: Classic histopathological and immunostaining features of IgG4-related disease
(A) Dense inflammatory infiltrate rich in IgG4-positive plasma cells (staining brown) in a prostate biopsy from a patient with IgG4-related disease of the prostate gland. (B) Lymphoplasmacytic infiltrate and storiform fibrosis in a prostate gland biopsy. Plasma cells are enmeshed by an irregularly whorled fibrotic process called storiform fibrosis.

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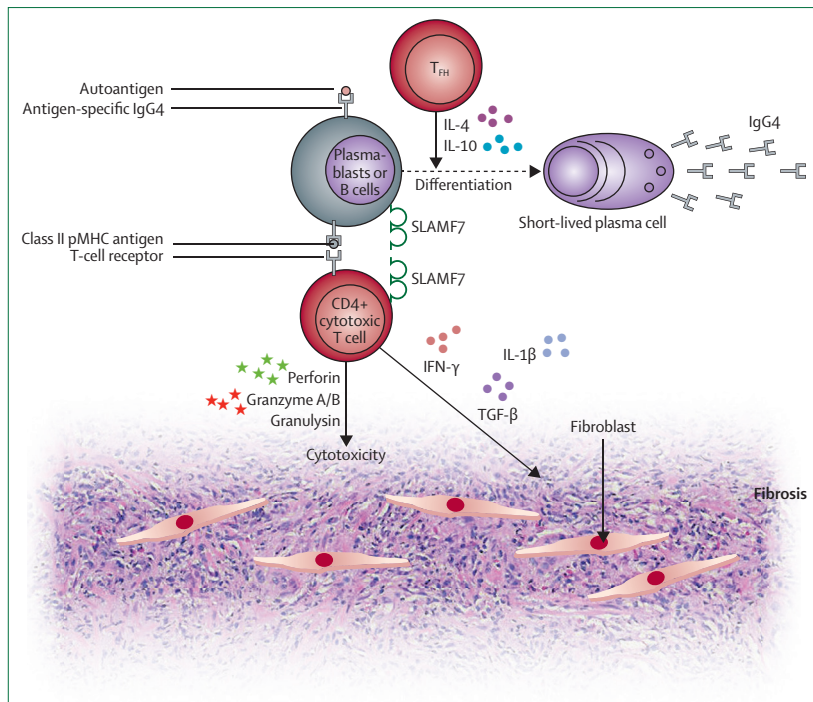


Figure 2: Pathophysiology of IgG4-related disease

A central event in disease pathophysiology is the presentation of autoantigen by plasmablasts or B cells to CD4 cytotoxic T cells (CD4 CTLs). Both CD4 CTLs and the antigen-presenting cells express SLAMF7, which makes SLAMF7 an appealing therapeutic target. CD4 CTLs express cytotoxic molecules, including perforin and granzymes, which contribute to cell death. CD4 CTLs produce profibrotic proteins such as IFN- γ , IL-1 β , and TGF- β . The class switch to IgG4 production by plasma cells appears to be driven by T_H cells via the secretion of IL-4 and IL-10. IL=interleukin. SLAMF7=signalling lymphocytic activation molecule F7. IFN=interferon. TGF=transforming growth factor. pMHC=peptide-major histocompatibility complex. T_H=follicular T-helper cells.

cell and neutrophilic infiltrates are highly atypical of the disease.

IgG4-related disease involves antigen-driven interactions among B-cell and CD4 T-cell lineages. Two types of CD4 T cells are involved in the disease: CD4 cytotoxic T lymphocytes (CD4 CTLs) and CD4 follicular T-helper (T_H) cells.¹¹ Of these, CD4 CTLs appear to be a disease linchpin; their interaction with B cells is essential to drive the disease and its associated fibrotic processes.¹²

CD4 CTLs often represent about 80% of all infiltrating CD4 T cells in tissues from patients with IgG4-related disease. These CD4 CTLs express a surface transmembrane molecule known as SLAMF7 (signalling lymphocytic activation molecule F7), which is involved in cytotoxicity, humoral immunity, autoimmunity, cell survival, cell adhesion, and lymphocyte development.¹³ The circulating counterparts of these tissue CD4 CTLs are highly clonally-restricted and express multiple profibrotic molecules, including interleukin (IL)-1 β , interferon γ , and transforming growth factor β . CD4 CTLs can contribute to fibrosis in situ by multiple mechanisms, including the secretion of profibrotic cytokines and the induction of apoptosis in targeted cells (figure 2).^{11,12}

In patients with IgG4-related disease, B-cell depletion leads to profound, swift clinical responses but a slow

decline of CD4 CTLs. A central hypothesis regarding the pathophysiology of disease is that activated B cells present antigen to and drive the activation of CD4 CTLs at sites of disease.¹¹ SLAMF7 is also expressed on plasmablasts and some double-negative (eg, CD19-positive, IgD-negative, and CD27-negative) B cells, which also express MHC class II molecules and thus might present antigens to CD4 CTLs, as observed in lupus.¹⁴ T_H cells appear to drive the production of IgG4 by B cells via the secretion of IL-4 and IL-10 (figure 2).¹⁵ Circulating subsets of T_H cells can also promote antibody production by B cells in vitro.¹⁶

There has also been growing interest in the role of the innate immune system in IgG4-related disease.¹⁷ Tissue-resident (type 2) macrophages, plasmacytoid dendritic cells, and eosinophils are recognised to participate in the pathophysiology of the disease, but a comprehensive discussion of these aspects of disease pathophysiology is beyond the scope of this review.

Little is known about the antigens that instigate disease. Galectin-3 has been identified as a major potential autoantigen in IgG4-related disease.¹⁸ In a cohort of 121 patients with IgG4-related disease and multi-organ involvement,¹⁸ autoantibodies against galectin-3 were found in 28% of patients. Presence of these autoantibodies correlated strongly with elevated plasma concentrations of galectin-3, suggesting a breach of tolerance driven by the accumulation of this endogenous protein. Other potential autoantigens have been identified, including annexin-A11,¹⁹ laminin-511,²⁰ and prohibitin.²¹ However, it is unlikely that a single antigen alone is responsible for triggering IgG4-related disease.

Clinical characteristics of disease

IgG4-related disease affects nearly every organ, but particularly the meninges, orbits (extraocular muscles and retrobulbar masses), lacrimal glands, salivary glands, thyroid gland, pancreas, bile ducts, lungs, kidneys, aorta, and retroperitoneum. Approximately 40% of patients present with clinically-evident disease in a single organ, but presentations involving five or six organs are not uncommon. Moreover, the disease can evolve over time, with new organs becoming involved in a metachronous manner. IgG4-related disease is typically a slowly progressive disease that is often present for months or even years before a diagnosis is established.

Weight loss of 5–15 kg is common in patients with IgG4-related disease. By contrast, fevers are distinctly unusual and nearly always implicate another diagnosis. Even patients with multi-organ involvement and substantial disease burden within an individual organ (eg, the lungs) can remain asymptomatic for long periods of time. This characteristic means that irreversible organ injury can occur even before the patient is aware of having the disease.

Diagnostic delay and the late institution of treatment can lead to major organ dysfunction and failure (panel 1). The pancreas is perhaps the organ injured most commonly.

Panel 1: Consequences to each organ system of not diagnosing and treating IgG4-related disease

Meninges

- Varied neurological complications depending on site of involvement (eg, cranial nerve palsies from basilar meningitis)

Posterior pituitary

- Anterior pituitary failure resulting in deficiencies of thyroid, adrenal, and sex hormones
- Posterior pituitary failure resulting in diabetes insipidus

Orbits and cavernous sinus

- Orbital pseudotumor
- Proptosis
- Vision loss through compression of optic nerve
- Cranial nerve palsies

Extraocular muscles (part of the orbits)

- Inhibition of full extraocular movements (eg, diplopia and limited field of vision)
- Proptosis

Sinuses and nose

- Anosmia
- Loss of taste
- Destructive bone lesions resembling granulomatosis with polyangiitis or malignancy

Lacrimal gland

- Sicca features

Major salivary gland

- Sicca features

Thyroid gland

- Riedel's thyroiditis, associated with a locally-invasive lesion that can threaten airway compromise and damage other regional structures

Lungs

- Pulmonary failure through interstitial lung disease

Pleura

- Thickened, fibrotic pleura, which can require decortication

Pericardium

- Constrictive pericarditis

Mediastinum

- Fibrosing mediastinitis with compromise of local structures (eg, superior vena cava, pulmonary arteries and veins, and mainstem bronchi)

Aorta

- Thoracic aortic aneurysm
- Inflammatory abdominal aortic aneurysm

Kidneys

- Renal failure secondary to tubulointerstitial nephritis
- Nephrotic syndrome caused by IgG4-related membranous glomerulonephropathy
- Unilateral or bilateral renal atrophy due to hydronephrosis

Pancreas

- Exocrine pancreatic failure, leading to poor nutrition and major weight loss (9–23 kg)
- Endocrine pancreatic failure leading to diabetes

Biliary tract

- Chronic hepatobiliary failure

Retroperitoneum

- Chronic pain
- Hydronephrosis, which can lead to renal injury

Mesentery

- Chronic pain
- Interference with major vessels (eg, inferior vena cava)

Substantial weight loss in a patient with IgG4-related disease usually implicates an injury to the exocrine pancreas, impairing the patient's ability to produce digestive enzymes and therefore to absorb nutrients. Such patients often have low serum amylase or lipase concentrations, reflective of burned-out autoimmune pancreatitis. The diagnosis of exocrine pancreatic insufficiency can be confirmed by the finding of low concentrations of elastase in stool. Management requires dietary supplementation with pancreatic replacement enzymes. Endocrine pancreatic insufficiency in IgG4-related disease leads to diabetes. A list of the common end-organ complications of IgG4-related disease is shown in panel 1.

Disease classification and subsets

Classifying IgG4-related disease into distinct subsets is intuitively appealing because the identification of disease subsets and their corresponding biological features might

lead to insights that guide treatment. The ultimate utility of identifying disease subsets is to discover meaningful biological differences between subsets that offer insights into prognosis or guide therapy.

The American College of Rheumatology and European League Against Rheumatism (ACR/EULAR) have produced a classification criteria for IgG4-related disease.²² These criteria were designed to identify patients with IgG4-related disease for enrollment in research studies, not for the identification of disease subsets. However, using data from 765 patients with IgG4-related disease from the patient cohort used to develop the ACR/EULAR criteria, one study²³ categorised the disease into one of four phenotypic groups by latent class analysis of the distribution of the involved organs. These groups were: (1) pancreato-hepatobiliary disease (31%), (2) retroperitoneal fibrosis with or without aortitis (24%), (3) head and neck-limited disease (24%), and (4) classic Mikulicz

	Proliferative subtype	Fibrotic subtype
Origin of disease	Glandular tissues, epithelial tissues	Extraglandular tissues, sometimes body regions (eg, retroperitoneum, mediastinum) rather than distinct organs
History of atopy	High frequency	Low frequency
Number of organs affected	Commonly multiorgan	Commonly single organ or single-body region
Organs affected	Lymph nodes, lacrimal gland, major salivary glands, pancreas, bile ducts, kidney, lung, pituitary, paranasal sinus	Retroperitoneum, aorta and periaortic tissue, mesentery, mediastinum, pachymeninges, thyroid (Riedel's)
Laboratory parameters		
IgG4 level	High	Normal or slightly elevated
IgG1 level	High	Normal or slightly elevated
IgE level	High	Normal or slightly elevated
Eosinophilia	Common	Unusual
Hypocomplementemia	Common	Unusual
Potential autoantigens	Laminin-511, galectin-3, annexin A11, prohibitin	Undefined
Pathology		
Lymphoplasmacytic infiltrate	Dense: >50, often >100; IgG4/IgG ratio >40%	Can be sparse; fewer than in proliferative subtype, but IgG4/IgG ratio >40%
IgG4-positive cells per HPF		
Germinal centre	Common	Unusual
Storiform fibrosis	Common	Common
Obliterative phlebitis	Occasional	Common
Responsiveness to treatment	Excellent	Less robust than in proliferative subtype, but good if initiated early

HPF=high-power field.

Table: Characteristics of the proliferative and fibrotic IgG4-related disease subtypes

syndrome with systemic involvement (22%). These groupings had substantial overlap with regard to organ system involvement across phenotypes. Nevertheless, there were important demographic differences among the groups with regard to certain features. Patients in group three (head and neck-limited disease) were far more likely to be female and Asian than were patients in the other groups. Patients in group four (Mikulicz syndrome and systemic involvement) had the highest median serum IgG4 concentrations.

An intuitive approach to classifying IgG4-related disease on the basis of a patient's clinicopathological characteristics is to differentiate two overlapping subtypes of disease: proliferative and fibrotic (table). Separation into these subsets appears to have some relevance to broad disease phenotype and responsiveness to therapy, although the biological features that separate these subtypes remain unknown.

In patients with the proliferative type of IgG4-related disease, the glandular and epithelial tissues are primarily affected. Common manifestations in this subset include lymphadenopathy, dacryoadenitis, sialadenitis (appendix p 1), autoimmune pancreatitis (appendix pp 2–3), IgG4-related sclerosing cholangitis, lung disease, tubulointerstitial nephritis, paranasal sinusitis, and hypophysitis. Patients with proliferative IgG4-related disease tend to have high serum concentrations of IgG4, IgG1 and IgE, hypocomplementemia, peripheral eosinophilia, and multi-organ disease often accompanied by atopy.^{24,25}

By contrast, the fibrotic type of the disease usually involves extra-glandular sites and can involve a body region rather than a specific organ. Examples of patients in this subset include those with retroperitoneal fibrosis (appendix pp 3–4), sclerosing mesenteritis, and fibrosing mediastinitis, although organs such as the thyroid gland (Riedel's thyroiditis) and meninges (hypertrophic pachymeningitis) can also be involved.

Pathology findings in patients with fibrotic disease include advanced stages of fibrosis, sparse lymphoplasmacytic infiltrates, and fewer germinal centers than in patients with proliferative disease.¹⁰ Disease affecting a single organ or body region is common^{26,27} in patients with the fibrotic type of disease, who often have normal or only mildly elevated serum IgG1, IgG4, and IgE concentrations. These patients are seldom hypocomplementemic, rarely have a peripheral eosinophilia, and are less likely to show atopy than patients with proliferative disease. To date, most of the work toward identifying disease-relevant autoantigens has been done in patients classified most appropriately as having proliferative disease.

The proliferative versus fibrotic classification construct has limitations due to overlap between these two subsets. Fibrosis occurs within the organs of patients with the proliferative subtype of disease, and some patients who fit into the fibrotic subtype also have organ involvement that is compatible with the proliferative subtype. For example, some patients with retroperitoneal fibrosis also have IgG4-related sialoadenitis of the submandibular glands. It is conceivable that the proliferative and fibrotic subtypes

See Online for appendix

represent different stages of the same inflammatory process, and that the proliferative stage is less easily identified in patients with organ involvement initially in the retroperitoneum, mesentery, or mediastinum and is therefore often overlooked. However, it is more likely that the phenotypic presentations now categorised as proliferative or fibrotic involve determinants that remain unrecognised at the present time. For example, responses to injury might vary across organ type or body region via mechanisms still poorly understood, accounting to some degree for differences in the presentation of these two subsets.

Treatment approaches

The goal of disease management in patients with IgG4-related disease is to reduce inflammation, maintain disease remission, and preserve organ function while minimising the adverse effects of treatment. All patients with active, symptomatic IgG4-related disease require treatment, and some require it urgently. Patients who are asymptomatic yet show signs of disease progression in vital organs also require treatment.^{9,28}

Watchful waiting is appropriate in a minority of patients.²⁹ A review³⁰ of therapeutic strategies from Japan suggested that some asymptomatic patients with IgG4-related disease with single-organ involvement, such as dacryoadenitis or sialoadenitis, might be monitored without treatment. However, long-term follow-up for such patients is lacking. The first international consensus guidance statement on the management of IgG4-related disease declared that not all manifestations of disease require immediate treatment and proposed watchful waiting as an appropriate strategy for patients with asymptomatic lymphadenopathy or mild submandibular gland enlargement.²⁸

Clinical experience suggests that although disease occasionally recedes temporarily in patients with single-organ disease, these patients can eventually have disease recurrences in major organs, often months or even years later, resulting in serious organ involvement or injury that could have been prevented with proper surveillance.^{28,31–34} The indolent nature of IgG4-related disease suggests that in patients for whom watchful waiting is selected, evaluations approximately every 6 months might be appropriate. Surveillance consists of educating patients to report new symptoms and to obtain regular laboratory evaluations, typically including a complete blood count, serum chemistries, IgG subclasses, IgE (if IgE is elevated at baseline), and complement components C3 and C4. Intermittent imaging can also be important in some patient subsets (eg, those with retroperitoneal fibrosis).

Remission induction

Successful remission induction is defined as the resolution of symptoms related to active disease and normalisation (or substantial improvement) of most biochemical

and radiological abnormalities.³⁵ IgG4-related disease—particularly the proliferative subtype—is highly responsive to treatment. Patients typically exhibit signs of improvement within days to several weeks of beginning treatment, depending on the type of organs involved.

Glucocorticoids are the cornerstone of treatment for most patients with IgG4-related disease, and patients typically respond well to this intervention.^{9,36–38} Indeed, failure to respond to an appropriate dose of glucocorticoid therapy constitutes an exclusion criterion in the ACR/EULAR Classification Criteria for IgG4-related disease.²² In countries where access to biological agents remains limited due to cost, glucocorticoids are used nearly exclusively or in combination with a conventional disease-modifying anti-rheumatic drugs (DMARDs). DMARDs are used with the intent to reduce glucocorticoid toxicity, although there is little evidence for their efficacy in that regard. The goal of induction therapy at many centres is to discontinue glucocorticoids within 3–6 months, but treatment strategies vary across countries.^{28,35,39}

For patients with multi-organ disease and high serum IgG4 concentrations, remissions are unlikely to be induced without glucocorticoid courses lengthy enough to cause substantial toxicity. The mean age at diagnosis of IgG4-related disease is approximately 60 years; patients consequently often have substantial comorbidities (eg, diabetes, obesity, osteoporosis, and hypertension). These comorbidities comprise major contraindications to prolonged glucocorticoid therapy, particularly as IgG4-related disease often has a substantial effect on the pancreas. A biological agent—particularly a B-cell depleting medication—should be considered if available.

Maintenance therapy

Whether long-term maintenance therapy is needed for all patients with IgG4-related disease remains a debate.²⁸ The disease is prone to recurrence, the risk of which increases with the number of organs affected at baseline and the baseline serum IgG4 concentration, among other factors. It is difficult to suggest a remission maintenance strategy that is applicable to all patients. Practices with regard to remission maintenance strategies differ to some degree from country to country but can be individualised according to a patient's specific disease features, extent of disease-related damage, baseline comorbidities, and previous treatment responses, including the durations of earlier treatment-induced remissions.

Kamisawa and colleagues⁴⁰ have suggested long-term maintenance therapy with low-dose prednisone for up to 3 years. This strategy is common in many Asian countries, where patients are generally maintained on doses of prednisone ranging from 5 to 10 mg per day.^{30,41–44} By contrast, clinicians in Western countries have advocated shorter prednisone courses,⁴⁵ with low-dose maintenance prednisone considered for patients with refractory disease or for those whose disease recurs shortly after prednisone cessation.³¹

Initial prednisone dose and adverse effects of glucocorticoids

An initial dose of 0·6 mg/kg per day (30–40 mg per day) of prednisone is recommended for autoimmune pancreatitis in Japan,^{32,37,40,46} with a nationwide survey showing no significant difference in effect between 30 mg per day and 40 mg per day.³² An international consensus guidance document concurred that prednisone at a dosage of 30–40 mg per day is appropriate for initial treatment.²⁸ In practice, the initial dosage of glucocorticoids varies widely, and some authors have reported the use of higher doses in patients with severe complications (pancreatic, pulmonary, renal and retroperitoneal involvement) but lower dosages in patients aged 70 years or older or those with comorbidities.

A randomised controlled trial⁴⁷ in China evaluated the effect of a high-dose prednisone regimen (0·8–1·0 mg/kg per day) compared with a medium-dose regimen (0·5–0·6 mg/kg per day). Patients with more organ involvement and a higher IgG4-related disease responder index⁴⁸ score were more likely to relapse on the medium-dose regimen. An aggressive approach in patients prone to recurrence might therefore be appropriate, particularly if there are few options for concomitant use of biologicals, but lower starting doses of prednisone are wise in the setting of less severe disease.³¹ When vital organs are involved, aggressive treatment is needed because of the risk of serious organ dysfunction and failure of the pancreas, liver, kidneys, lungs, and other organs.⁹

The starting dose of prednisone is usually maintained for 2–4 weeks.^{28,40,46} At some centres,^{45,49} treatment is initiated with prednisone at 40 mg per day and this dose is maintained for 4 weeks. Thereafter, the prednisone dose is often tapered by 5 mg per day for 1–2 weeks,^{28,32,35,40,50} guided by the rapidity of clinical improvement, changes in serum biomarkers (eg, bilirubin, hepatic aminotransferases, and serum IgG4 concentrations), and follow-up imaging studies.^{28,40} Modifications of this strategy have been offered.³⁷

A prospective, multicentre trial in Japan,⁴³ comprising 61 patients, investigated adverse events during glucocorticoid treatment in patients with IgG4-related disease. The authors reported glucose intolerance in 25 (41%) of 61 patients, dyslipidemia in 16 (26%), and major infections in 11 (18%), despite a maintenance dose of only 5–10 mg per day in most patients.⁴³ Crucial concepts in management include careful monitoring for glucocorticoid toxicity and proactive interventions to limit such toxicity, particularly by reducing treatment duration and using effective glucocorticoid-sparing agents when available.⁵¹ Reported approaches to glucocorticoid therapy in patients with IgG4-related disease are outlined in the appendix p 5.

Conventional DMARDs

Azathioprine, mycophenolate mofetil, methotrexate, leflunomide, and cyclophosphamide are commonly used worldwide in combination with glucocorticoids in patients

with IgG4-related disease.²⁸ Despite the frequency with which they are used, however, there exists scant evidence that these agents add significantly to the efficacy of glucocorticoids, and most data derive from retrospective studies.

A prospective cohort study from China,⁴¹ comprising 104 patients with newly-diagnosed IgG4-related disease compared glucocorticoid monotherapy with the combination of glucocorticoids and cyclophosphamide (50–100 mg per day). Both treatment regimens were equally effective in inducing remission, but the addition of cyclophosphamide lowered the recurrence rate over a 12-month follow-up. It is also important to note the high frequency of disease flares in patients treated with glucocorticoid monotherapy: 12 (23%) of 52 patients in this group had clinical relapses within 1 year despite the use of maintenance glucocorticoids, and an additional 8 (15%) patients had serological relapses in the absence of worsening clinical symptoms or imaging results.⁴¹

Even if the addition of cyclophosphamide reduces the incidence of disease flare, the side-effect profile of this drug is a disincentive to its widespread use, except for patients with severe disease for whom B-cell depletion therapy is not available, or those for whom the avoidance of additional glucocorticoid toxicity is paramount. A summary of the evidence from studies of conventional DMARDs in IgG4-related disease is provided in the appendix p 6.

Biological agents

The pathophysiological understanding of IgG4-related disease indicates logical targets for therapy directed against B cells or CD4 T cells, specifically CD4 CTLs expressing SLAMF7. Patients with IgG4-related disease can often be treated with a biological agent alone, without the need for concomitant therapies that complicate the understanding of efficacy, add potential toxicity, and confound the interpretation of mechanistic studies.

B-cell depletion with rituximab was first used in patients with IgG4-related disease because of a search for a potentially effective glucocorticoid-sparing medications,⁵² and because the elevated concentrations of IgG4 seen in most patients provided evidence for a role of humoral immunity. B-cell depletion likely operates through several mechanisms. First, B-cell depletion interrupts antigen presentation to CD4 CTLs; indeed, the concentration of CD4 CTLs in the peripheral blood and in tissue declines following targeted B-cell depletion.^{12,53}

Second, B-cell depletion decreases tissue fibrosis⁵⁴ by reducing the number of infiltrating, activated myofibroblasts. Decreased cytokine production by B cells is also likely to have a positive effect on disease activity. Third, the reduction of serum IgG4 concentrations is beneficial, particularly in patients with dramatic elevations of serum IgG4 concentrations who might be prone to developing immune complexes that can contribute to tissue injury.

Evidence for the efficacy of rituximab was shown in a trial of 30 patients with IgG4-related disease,⁵⁵ with disease responses occurring in 29 (97%) patients. 23 (77%) patients achieved the primary outcome of remission, and 12 (40%) patients remained in complete remission at 12 months, despite the fact that rituximab was not re-dosed during follow-up. However, in a multicentre nationwide study⁵⁶ done in France assessing long-term safety and efficacy of rituximab in 33 patients over 2 years, only 17 (52%) patients achieved glucocorticoid withdrawal, and 13 (42%) patients experienced disease relapses. Moreover, serious infections or hypogammaglobulinemia were observed in 11 (33%) patients,⁵⁶ further underscoring the need for new treatment approaches.

Other treatment strategies directed against B cells also appear promising. A phase 2 trial of XmAb5871,⁵⁷ a monoclonal antibody directed against both FcγRIIb and CD19, elicited positive treatment responses in 12 of 15 patients, with disease control obtained in the absence of glucocorticoids in most subjects. Abatacept (anti-CTLA4-Ig), which interferes with T cell co-stimulation, was reportedly effective in one anecdotal report⁵⁸ and is now being tested in a small, open-label trial.

SLAMF7 is a highly appealing drug target because it is present on both CD4 CTLs and B cells. Depletion of activated cells that express this surface molecule would therefore directly address the interaction between CD4 CTLs and the antigen-presenting B cells that is felt to be fundamental to IgG4-related disease. A proof of concept trial with the anti-SLAMF7 monoclonal antibody elotuzumab, used to treat relapsed multiple myeloma, is under development.

Biomarkers

Studies in IgG4-related disease benefit from a wealth of good potential biomarkers, some of which are useful in clinical management. Serum IgG4 concentrations—which are elevated in most patients—have proven particularly useful. In many patients with elevated serum IgG4 concentrations at baseline, trends in IgG4 concentrations over time constitute a reliable biomarker of disease activity and an indicator of when re-treatment should be considered. Moreover, serum IgG4 concentrations reflect the extent of disease and the number of organs affected, correlating well with IgG4-related disease responder index scores.⁵⁹ Perhaps most importantly, changes in serum IgG4 concentrations correspond to treatment responses, decreasing after effective treatment with either glucocorticoids or rituximab.^{59,60} Steady elevations of serum IgG4 concentrations predict the development of clinical disease flares.⁶¹

The decline in serum IgG4 concentrations following effective treatment probably reflects control of the primary inflammatory process, permitting a corresponding decrease in the counterregulatory response represented by IgG4 elevation. Serum IgG4 concentrations decline reliably with prednisone or rituximab treatment but often

do not return to completely normal concentrations for months or even years, presumably because of continued production of IgG4 by long-lived plasma cells in the bone marrow. A study³² reported that 115 (63%) of 182 of patients with IgG4-related disease still maintained IgG4 concentrations above the upper limit after treatment with glucocorticoids and the achievement of disease remission.

Elevated serum IgE concentrations are present in 50–90% of patients with IgG4-related disease, often to a striking degree.⁶² Therefore, both serum IgE and IgG4 are frequently elevated, presumably under the control of cytokines. IgE production is likely driven by IL-4 and IgG4 by the combination of IL-4 and IL-10 (figure 2), but this supposition needs to be confirmed. A small number of patients with IgG4-related disease have dramatic increases in serum IgE but normal IgG4 concentrations. The converse is also true. Serum IgE concentrations correlate with disease activity in many patients and, in some, track more closely with disease activity than do IgG4 concentrations.^{62,63} A full understanding of the biological basis of serum IgE elevation remains to be elucidated, but for many patients this represents a useful biomarker of treatment response.

Serum complement C3 and C4 concentrations are often inversely correlated with IgG4 and IgE concentrations in the setting of active disease, declining as disease activity worsens. This hypocomplementemia suggests the possibility of a role for immune complexes in pathophysiology in some patients (appendix p 8). C3 and C4 are decreased in the serum in approximately a third of patients with autoimmune pancreatitis, and hypocomplementemia appears to correspond particularly well to the presence of IgG4-related tubulointerstitial nephritis. However, low complement levels can also be observed in patients without overt renal disease.^{38,64}

IgG4 does not bind complement effectively in most circumstances, but a plausible explanation for the finding of hypocomplementemia is that IgG1 has a prominent role in activating complement via the classical pathway.^{64,65} Another potential explanation is that IgG4 activates the complement system through the mannose-binding lectin pathway.^{65,66} Altered glycosylation of IgG1 and IgG4 antibody subclasses might also have a role in causing hypocomplementemia in patients with IgG4-related disease.⁶⁷

Activated B cells, or plasmablasts, are often defined by the surface expression of various molecules: they express CD38, CD27, and CD19 (at low levels) but do not express CD20.⁶⁸ Circulating plasmablasts are increased in patients with IgG4-related disease compared with healthy controls, and also in patients with many other immune-mediated conditions and malignancies.⁶⁹ Concentration of these cells also correlates with serum IgG4 concentrations, the number of involved organs, and IgG4-related disease risk index.⁷⁰ Before patients undergo treatment, quantification of plasmablasts is more sensitive than is the measurement of serum IgG4 concentrations for the diagnosis of IgG4-related disease. However, clinical experience indicates that after treatment, the value of plasmablast

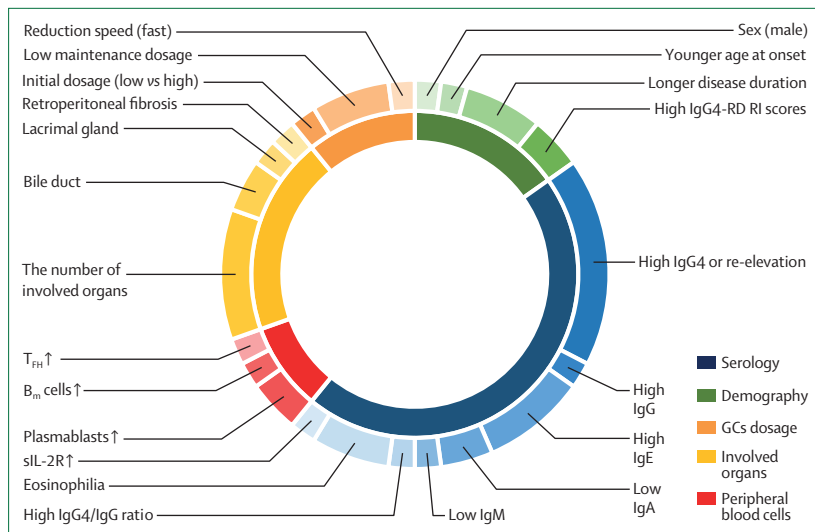


Figure 3: Risk factors for flare in IgG4-related disease

Risk factors are divided by category: serology (blue), demography (green), glucocorticoid dosage (orange), organ involvement (yellow), and concentrations of peripheral blood cells (red). The length of each bar indicates the relative frequency of reference to each risk factor in the published literature. RI=Responder Index. B_m =memory B cells. T_{FH} =follicular T-helper cells. GCs=glucocorticoids. sIL-2R=soluble interleukin-2 receptor.

Panel 2: Controversies and areas of intense research interest

- What is the optimal glucocorticoid regimen for patients with IgG4-related disease?
- Are there patients who should forgo glucocorticoid treatment in favour of a biological treatment approach (eg, B-cell depletion)?
- What treatment approaches might be effective for the fibrotic manifestations of IgG4-related disease, and how should physicians employ such strategies in the context of other treatments?
- How should patients with the fibrotic subset of IgG4-related disease be managed?
- When and how should patients be re-treated?
- How can current or investigational imaging approaches be used to monitor disease activity on a longitudinal basis?
- Is there a core set or finite group of specific antigens that trigger B-cell responses in IgG4-related disease?
- Do the same antigens that trigger B-cell responses also activate CD4 CTLs?
- What is the role of complement in IgG4-related disease? Why are some patients profoundly hypocomplementemic, whereas others have no evidence of complement activation?
- What is the extent of genetic contributions to IgG4-related disease risk?
- Do activated B cells and CD4 CTLs contribute to tissue fibrosis?
- Do activated B cells present antigen to CD4 CTLs in the tissue?
- What molecules determine IgG4 vs IgE class-switching?
- What are the therapeutic effects of direct targeting of T_{FH} or CD4 CTL?
- Is there a relationship between malignancy and IgG4-related disease?

measurement declines. In addition, flow cytometry assays for plasmablasts must be done locally, and measurement techniques vary widely across centres, thus circumscribing the general utility of these assays.

CD4 CTL count is also a potential biomarker,^{12,71} but the utility of enumerating these cells as part of the longitudinal management of patients' needs to be confirmed in a prospective study. Such assays will be subject to the

same limitations that apply to flow cytometry studies on B cells. Several groups have investigated the potential of type 2 T_{FH} cells as biomarkers of IgG4-related disease.^{72,73} Type 2 T_{FH} cells correlate with serum IgG4 and IgE concentrations, circulating plasmablasts, and the IgG4-related disease risk index. These cells decline in the peripheral blood following glucocorticoid treatment.

Risk factors for relapse

Risk factors for disease relapse can be divided into five main categories: demographic features, organ involvement, serological findings, medications, and changes in peripheral blood cell populations. Reported risk factors for relapse are shown in figure 3, organised according to their frequency as reported in the literature.

Being male has been implicated as a risk factor for disease flare in some studies, with a hazard ratio of 3.14 ($p=0.003$) reported in one study.⁷⁴ This is interesting in view of the moderate male predilection for the disease overall. The number of organs involved at baseline also appears to be a risk factor for disease relapse.^{39,41,44,61} The number of organs affected by the disease correlates highly with serum IgG4 concentrations, and patients with higher serum IgG4 concentrations at baseline are more likely to relapse.^{41,44,61,63} This is also true for patients with higher serum IgE or eosinophil concentrations at baseline.^{62,63}

Tapering or discontinuing therapy heightens the risk of flare. A randomised controlled trial⁴⁴ of patients with autoimmune pancreatitis showed that the maintenance dose of prednisone was lower in patients who relapsed (mean 6.5 [SD] 2.5 mg per day) than in those who did not relapse (8.2, [SD] 1.9 mg per day). Similarly, a prednisone maintenance regimen of less than 5 mg per day was strongly associated with disease relapse.⁷⁵

Special situations

In selected situations, surgical treatment or other mechanical interventions (eg, stent placement) are complementary approaches to medical therapy. Surgical biliary drainage, now rare, was common before IgG4-related disease was recognised as a specific disease entity. Surgical resection of some organs that are severely damaged is occasionally necessary. The placement of biliary stents, ureteral stents, or nephrostomy tubes are sometimes required at the start of medical therapy, but ideally, the use of these measures is temporary. Removal of the stents or tubes at the earliest possible opportunity is important from the perspective of patient comfort and the avoidance of complications.

IgG4-related retroperitoneal fibrosis

The ureters course medially as they traverse from the renal pelvis to the bladder. Near the bifurcation of the iliac arteries from the distal aorta, a site often affected by retroperitoneal fibrosis, the ureters often become entrapped by the retroperitoneal fibrosis and periaortitis, leading to hydronephrosis. The relief of obstruction is the first

priority in management of hydronephrosis. Ureteral stents or nephrostomy tubes might be indicated, depending on the anatomic localisation of the obstruction. Ureteric stenting, surgical ureterolysis, laparoscopic ureterolysis, and omental wrapping might all be required,^{31,76} and mild cases can be treated with immunosuppression alone as the first option.⁷⁷ Some patients with retroperitoneal fibrosis respond very well to glucocorticoids or rituximab, with prolonged treatment responses. Intermittent retreatment is often required.

Sclerosing mesenteritis

Sclerosing mesenteritis often occurs in a spectrum of disease with retroperitoneal fibrosis, with the occurrence of inflammation and fibrosis at the root of the mesentery. The process can be clinically silent until it involves crucial vascular structures or organs. Complex debulking surgery is often required at the time sclerosing mesenteritis is identified. Medical therapies for IgG4-related sclerosing mesenteritis are frequently ineffective in these patients, although such treatments can be administered with the goal of preventing further disease progression. Debulking surgery followed by medical therapy (eg, B-cell depletion) might be the best approach for many patients amenable to surgery.

Biliary tract disease

In patients with IgG4-related sclerosing cholangitis with obstructive jaundice, stent placement, and biliary drainage before the initiation of glucocorticoid therapy can be appropriate.⁷⁸ Stents are inserted to prevent or assist in the treatment of ascending cholangitis in the context of obstructive jaundice. The stents usually fall out or can be removed within 4 weeks of the initiation of immunosuppressive therapy. Infection and migration of biliary stents are recognised complications and need to be closely monitored.

Riedel's thyroiditis

Riedel's thyroiditis (appendix p 9) can cause a series of compression symptoms, such as hoarseness, dyspnea, and dysphagia. Although Riedel's thyroiditis arises in a gland, its histological features are often most consistent with the fibrotic subtype of IgG4-related disease, and therefore this lesion often responds poorly to immunosuppression. If detected reasonably early, Riedel's thyroiditis may be responsive to prednisone or rituximab.⁷⁹ Unfortunately, in many patients, the problem is not recognised until it has extended through the thyroid capsule and affected local tissues (eg, the trachea), at which time major surgery might be required.⁸⁰

Aortitis

IgG4-related aortitis can affect the abdominal and thoracic aorta, presenting as inflammatory aortic aneurysm in some patients. Surgeries such as arterial graft replacement, stent grafting, or endovascular repairs might be required

Search strategy and selection criteria

We based our review on a narrative rather than systematic literature search. We searched MEDLINE for papers published in the English language before March 31, 2019, on treatment and stratification of therapies in IgG4-related disease. We identified key abstracts from the European League Against Rheumatism and American College of Rheumatology annual meetings in 2016–18. Search terms were “IgG4-related disease”, “autoimmune pancreatitis”, “tubulointerstitial nephritis”, “complement”, “SLAMF7”, “B lymphocyte”, “pathology”, and “treatment”.

in IgG4-related vascular disease with aneurysm formation.⁸¹ As no surgical guidelines specific for inflammatory aortic aneurysm exist, standard management approaches for aneurysms are often appropriate.⁸² Timely use of immunosuppression can obviate the need for surgery if initiated early enough.

Future Steps

In less than 15 years, IgG4-related disease has gone from being an unrecognised disease to one that is recognised around the world. The disease has certain strong organ predilections, and clinicians from nearly every specialty now recognise the complications of IgG4-related disease within their fields. IgG4-related disease is one of the great mimickers of other diseases in the current era, and early recognition and treatment is crucial. Areas of uncertainty, ongoing controversy, and intense research interest are highlighted in panel 2.

The rapid achievement of insights into the pathophysiology of IgG4-related disease suggests the possibility of a variety of targeted treatment approaches. Looking ahead, a number of these approaches could profoundly alter the way in which IgG4-related disease is managed, shifting the focus away from glucocorticoids and non-biologic DMARDs to more specific therapies that can be selected and applied in a rational manner, taking into consideration each patient's clinical phenotype.

Contributors

JHS and WZ did the literature search, prepared the figures, and wrote and edited the manuscript. Both authors had final responsibility for submission of the manuscript.

Declaration of interests

JHS declares grant support from Roche/Genentech and Bristol-Meyers Squibb, and has received personal fees from Roche/Genentech. WZ declares no competing interests.

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